Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

Introduction to Model Selection
Reading • 5 min

Bayesian Model Selection
Reading • 30 min

Bayesian Model Selection Readings
Reading • 1h

Lesson 2: Cross Validation

Introduction to Cross-Validation
Reading • 25 min

K-Fold Cross-Validation
Reading • 2 min

Leave-One-Out Cross-Validation (LOOCV)
Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

Introduction to Optimization Techniques
Reading • 13 min

Optimization Using Gradient Descent
Reading • 40 min

Gradient Descent with Momentum
Reading • 45 min

Stochastic Gradient Descent (SGD)
Reading • 40 min

Constrained Optimization
Reading • 16 min

Lagrange Multipliers
Reading • 8 min

Convex Optimization
Reading • 8 min

Linear Programming
Reading • 15 min

Quadratic Programming
Reading • 10 min

Check Your Knowledge: Optimization Techniques
Graded Assignment • 15 min

Introduction to Model Selection

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Model selection is a crucial step in the machine learning pipeline, involving the choice of the best model from a set of candidate models based on their performance on a given dataset. The goal is to select a model that generalizes well to unseen data, avoiding both underfitting and overfitting.

Model Selection Criteria:

- **Generalization Performance:** The primary goal is to choose a model that performs well on new, unseen data, not just on the training data.
- **Complexity:** A model that is too complex may overfit the training data, capturing noise rather than the underlying pattern. Conversely, a model that is too simple may underfit, failing to capture important patterns.
- **Bias-Variance Tradeoff:** Balancing bias (error due to overly simplistic models) and variance (error due to overly complex models) is key to selecting an appropriate model.

Model Selection Strategies:

- **Train-Test Split:** The dataset is split into training and testing sets. The model is trained on the training set and evaluated on the test set. This approach is simple but might not give a complete picture of model performance.
- **Nested Cross-Validation:** Combines cross-validation with hyperparameter tuning in a nested structure to provide a more reliable estimate of model performance.
- **Information Criteria:** Criteria like Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) penalize model complexity and are used to select simpler models that perform well.

Overfitting and Underfitting:

- **Overfitting:** Occurs when the model performs exceptionally well on the training data but poorly on unseen data. It often happens with overly complex models.
- **Underfitting:** Happens when the model is too simple to capture the underlying patterns in the data, leading to poor performance on both training and unseen data.

Evaluation Metrics:

- **Accuracy:** The proportion of correct predictions (used for classification tasks).
- **Precision, Recall, and F1-Score:** Important metrics for imbalanced classification tasks.
- **Mean Squared Error (MSE) and R-squared:** Commonly used for regression tasks.
- **AUC-ROC:** Evaluates the performance of a binary classifier by measuring the area under the ROC curve.

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